PassFrame tuition

Coordinates & Transformations

Year 5 Accelerated — Mark Scheme — Lesson Worksheet & Homework (tutor copy)

Lesson Worksheet — Answers

Easy / Standard

1. -8

2. 11

3. (3, -13)

4. (-14, 4)

5. A — (5, 8)

6. The third quadrant (both x and y are negative).

7. The fourth quadrant (x is positive, y is negative).

8. (-7, -7)

9. (-18, 10)

10. A, B and D (P(-6, 14), Q(6, -14), S(6, 14)) — R is the only point where both coordinates are negative (the third quadrant)

11. (-13, 4)

12. A — (8, 8)

13. Midpoint = ((-9 + 11) / 2, 3) = (1, 3).

14. (-8, 7)

15. A — (0, -13)

Standard

1. (-14, -9)

2. (-16, -11)

3. (7, -6)

4. A(2, 2), B(2, 10), C(11, 10), D(11, 2).

5. A — (-13, 7)

6. Reflect in x-axis: (12, -9). Reflect in y-axis: (-12, -9). (This is the same as a 180° rotation about the origin.)

7. (19, -14)

8. Length = 13 - (-5) = 18 units. Midpoint = (-8, (-5 + 13) / 2) = (-8, 4).

9. (-8, -5)

10. A — (-17, -12)

11. (-22, 16)

12. A, B and C (Its width is 22 units, Its height is 14 units, The vertex (13, 10) is diagonally opposite (-9, -4)) — perimeter is actually 2 × (22 + 14) = 72 units, so the perimeter statement is false

13. (21, 21)

14. Reflect: (-5, -15). Translate 7 down: (-5, -22).

15. A — (7, -7)

Standard / Hard

1. (13, -6)

2. (8, 9)

3. (11, -15)

4. A — (-12, -5)

6. (18, -13)

7. Side length 14 units; perimeter = 4 × 14 = 56 units; centre at (0, 0).

8. (13, -8)

9. (-17, -21)

11. (-13, 11)

12. Reflect: (-9, -14). Rotate: (y, -x) = (-14, 9).

13. A — 60 units (width 16, height 14, perimeter = 2 × (16 + 14) = 60)

14. Subtract 4 from each x and 3 from each y: (-2, -2), (-2, 4), (2, -2).

5. (y, -x) = (-18, 0).

10. A, B, C and D (A becomes (0, 3), B becomes (9, 3), C becomes (9, -6), D becomes (0, -6))

15. 13 right and 13 up.

Homework — Answers

Easy / Standard

1. -14

2. 17

3. (4, -18)

4. (-15, 5)

5. A — (4, 12)

6. The third quadrant (both x and y are negative).

7. The fourth quadrant (x is positive, y is negative).

8. (-9, -9)

9. (-20, 11)

10. A, B and D (P(-8, 19), Q(8, -19), S(8, 19)) — R is the only point where both coordinates are negative (the third quadrant)

11. (-17, 4)

12. A — (9, 11)

13. Midpoint = ((-12 + 14) / 2, 5) = (1, 5).

14. (-10, 8)

15. A — (0, -17)

Standard

1. (-19, -12)

2. (-21, -15)

3. (5, -4)

4. A(2, 2), B(2, 13), C(14, 13), D(14, 2).

5. A — (-17, 9)

6. Reflect in x-axis: (16, -11). Reflect in y-axis: (-16, -11).

7. (24, -19)

8. Length = 17 - (-7) = 24 units. Midpoint = (-10, (-7 + 17) / 2) = (-10, 5).

9. (-9, -6)

10. A — (-22, -15)

11. (-27, 20)

12. A, B and C (Its width is 27 units, Its height is 17 units, The vertex (16, 12) is diagonally opposite (-11, -5))

13. (25, 25)

14. Reflect: (-6, -18). Translate 9 down: (-6, -27).

15. A — (8, -8)

Standard / Advanced

1. Rotate: (y, -x) = (-9, -4). Translate 3 left: (-12, -4).

2. Width = 20, height = 14. Perimeter = 2 × (20 + 14) = 68 units, so it is LESS than 80 units.

3. Reflect in y-axis: (-14, -9) → (14, -9). Sum = 14 + (-9) = 5.

4. 17 left: 17 - 17 = 0, giving (0, -6), which lies on the y-axis.

5. Centre = midpoint of opposite corners = ((-6+10)/2, (-6+10)/2) = (2, 2). Since both coordinates are

6. (16, 9)

7. Width 20, height 10, perimeter = 2 × (20 + 10) = 60 units. Centre = (-2, 1).

8. (4, -11)

9. (-14, 17)

10. (21, 19)

11. B, C and D (B becomes (4, -11), C becomes (-7, -11), D becomes (-7, 0))

12. (-14, 11)

13. Reflect: (-13, -19). Rotate: (y, -x) = (-19, 13).

14. A — 76 units (width 20, height 18, perimeter = 2 × (20 + 18) = 76)

15. Subtract 5 from each x and 4 from each y: (-2, -2), (-2, 5), (3, -2).

positive, the centre IS in the first quadrant.